

Curve fitting

1. Find a parabola that *exactly* passes through the three points $(0, 4)$, $(1, 9)$, and $(4, -2)$ by constructing a linear equation and then solving it.
2. Various voltages v_j are applied to a resistor (a device contained in almost all electronic circuits) and the resulting currents i_j are measured:

v [V]	i [mA]
0.20	27
0.40	49
0.60	64
0.80	85
1.0	108

- (a) The resistance R should satisfy Ohm's law, $v = iR$. However, there may be a systematic offset in your voltmeter. In order to take this into account, instead apply a least-squares fit to the model $v = v_{\text{off}} + iR$. Find both v_{off} and R , assuming that the voltage measurements are much more accurate than the current measurements. [Note that you might want to use a calculator or a computer here.]
- (b) Ohm's law is an idealization. Suppose that you want to check for corrections to this law by introducing a (large) "nonlinearity scale" in the form of a new constant v_{nlin} :

$$v + v^2/v_{\text{nlin}} = v_{\text{off}} + iR.$$

Explain how you could use fitting to try to measure the voltage scale v_{nlin} at which Ohm's law begins to fail (without actually applying very large voltages to the resistor). Which equations would you try to solve?